PORTFOLIO EXECUTIVE DECK

International Basketball Analytics

20 Years of FIBA Competitions (2005–2024): An End-to-End Data, ML & Decision Support System

Python + R + DuckDB + Parquet | 1,145 Games | 27,353 Player Records | 227 Automated Tests

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Open-Source Repository: https://github.com/miguelo0203/basketball-analytics

PROJECT AT A GLANCE

Executive Summary: From Raw Data to Court Decision

A complete data engineering and predictive pipeline built for professional basketball decision-makers.

01. Data Foundation

18 senior FIBA tournaments (EuroBasket, World Cup, Olympics) structured into an OLAP DuckDB warehouse with 12 normalized tables.

02. Advanced Metrics

Dean Oliver Four Factors normalized to pace, True Shooting %, and longitudinal career stability via Bayesian shrinkage.

03. Machine Learning

Chronological 17-fold walk-forward validation (1,105 test games). Brier Score 0.1967 (vs 0.2500 naive) and ECE 0.0314.

04. Decision Support

1.5-page pre-game coaching briefs, 6 objective player archetypes, and anti-hindsight validation mode.

Core Value: Reducing tactical uncertainty while preserving coaching intuition.

The Analytics Challenge in Professional Basketball

Why traditional boxscores and 40-page scouting dossiers fail modern coaching staffs.

The Status Quo: Information Overload

- Scouting reports are flooded with raw totals (PPG, RPG) that ignore pace and lineup context.
- Short-tournament variance (6-9 games) misleads decision-makers into over-valuing shooting hot streaks.
- Models trained with random k-fold suffer from temporal data leakage, learning future tactical eras.
- Analysts deliver disconnected charts instead of structured pre-game decision questions.

Our Solution: The Data-First Pipeline

- Pace-neutral Four Factors isolate true possession efficiency.
- Bayesian shrinkage (lambda = 0.75) stabilizes small-sample noise toward historical priors.
- Strict walk-forward expanding window guarantees zero future contamination.
- Concise 1.5-page coaching briefs delivering actionable tactical hypotheses before tip-off.

01

Data Architecture & Engineering

From Raw Historical Archives to In-Process Columnar OLAP with DuckDB and Parquet

DATA FOUNDATION

Historical Scope & Canonical Scale

Two decades of international men's senior basketball verified with SHA-256 cryptographic provenance.

1,145

OFFICIAL MATCHES

Every game across 18 FIBA tournaments (2005–2024).

18

TOURNAMENTS

EuroBasket (2005-2022), FIBA World Cup (2006-2023), Olympics (2008-2024).

27,353

PLAYER PERFORMANCES

Individual boxscore game logs across all 18 tournaments.

2,124

CANONICAL PLAYERS

Deterministic entity resolution mapping names across 20 years.

Integrity: 100% immutable data lake with zero external live API dependencies.

DATA ENGINEERING

Medallion Data Warehouse Architecture

Decoupled pipeline separating ingestion, relational validation, columnar storage, and analytics.

01_RAW

HTML/JSON Lake

18 tournament source archives stored for full offline reproducibility.

02_VALIDATED

DuckDB Warehouse

12 relational tables (28.5 MB uncompressed) with 3NF integrity.

03_ANALYTICS

Apache Parquet

11 high-performance columnar marts optimized for vectorized queries.

04_DELIVERY

Python & R APIs

Dual-engine consumption for ML training, statistical EDA, and briefs.

Performance: Sub-millisecond aggregations on 27k+ player rows directly in-memory.

DATA QUALITY

Automated Quality Assurance & Invariants

Rigorous mathematical contracts enforced prior to any statistical modeling.

200 Minutes / Game Rule

Every 40-minute regulation game strictly verifies exactly 200 player-minutes (plus 25 min per 5-min overtime). Any mismatch fails the pipeline.

Score & Boxscore Closure

Sum of points (2P + 3P + FT) is reconciled against official final match scores. Zero scoring discrepancies across 1,145 games.

Entity Resolution Engine

Unified 2,124 players across language variants, accents, and name changes (e.g. Dirk Nowitzki, Bojan vs Bogdan Bogdanovic) using custom slug hashing.

Quality Standard: Corrupt data yields corrupt decisions. We validate before we model.

02

Analytics & Modeling Engine

Pace Normalization, Longitudinal Stability, Player Archetypes, and Calibrated Machine Learning

METRICS FRAMEWORK

Pace Normalization & Dean Oliver Four Factors

Decomposing basketball efficiency into four fundamental, pace-neutral drivers.

Effective FG% (eFG%) (Shooting Efficiency)

Weights 3-pointers at 1.5x. Accounts for shot quality differences.

eFG% = (FGM + 0.5 * 3PM) / FGA

Turnover Rate (TOV%) (Ball Security)

Measures possessions ending in turnover per 100 plays.

TOV% = TOV / (FGA + 0.44 * FTA + TOV)

Off. Rebounding (ORB%) (Second Chances)

Percentage of available offensive rebounds secured.

ORB% = ORB / (ORB + Opp_DRB)

Free Throw Rate (FTR) (Foul Drawing)

Ratio of free throw attempts to field goal attempts.

FTR = FTA / FGA

Pace Neutrality: Comparing 65-possession games directly against 85-possession games.

Overcoming Small-Sample Tournament Noise

Using Bayesian shrinkage (lambda = 0.75) and cluster bootstrap to separate luck from skill.

The Small-Sample Trap

In a 7-game tournament, a 15/30 3-point shooter registers 50%. A swing of just 3 shots drops him to 40%. Evaluating players on raw short-term percentages produces severe tactical misjudgments.

Bayesian Shrinkage (lambda = 0.75)

We contract short-sample shooting metrics toward the player's historical career prior weighted by possession volume. High-volume players shrink less; low-volume samples regress toward baseline.

Cluster Bootstrap (B = 5,000)

We generate 95% non-parametric confidence bands across 3,767 qualified player campaigns (>= 40 min), providing coaches with realistic uncertainty intervals.

Longitudinal Asset: Tracking true shooting evolution across legends (Gasol, Nowitzki, Bogdanovic).

PLAYER TAXONOMY

6 Objective Functional Archetypes

Clustering 3,767 player campaigns via K-Means++ and PCA on 14 normalized per-40 metrics.

Primary Initiator

High usage, primary P&R handler, elite shot creation and passing volume.

Floor Spacer

High 3P volume, catch-and-shoot specialist, low interior touches, elite spacing.

Interior Hub

Post playmaker, rim finisher, high offensive rebounding and short-roll passer.

Floor General

Pure playmaker, exceptional assist-to-turnover ratio, low turnover rate, tempo controller.

Defensive Anchor

Rim protector, elite defensive rebounder, screen-and-roll finisher, low usage.

Balanced Wing

Versatile two-way wing, secondary creator, switchable defender, balanced shot profile.

Scouting Value: Moving beyond rigid 1-5 positions to evaluate true roster complement.

PREDICTIVE MODELING

Calibrated Machine Learning: LightGBM

Probabilistic match forecasting regularized with L2 penalty and isotonic calibration.

0.1967

BRIER SCORE

vs 0.2500 naive baseline (+21.3% quadratic error reduction).

0.0314

CALIBRATION ERROR (ECE)

Expected Calibration Error: within 3.14% of perfect calibration.

11.74

POINT MARGIN MAE

Mean Absolute Error in point spread across 1,105 test matches.

Honest Probability: An event assigned 70% probability occurs exactly 7 out of 10 times.

17-Fold Temporal Walk-Forward Protocol

Strict expanding-window splits preventing temporal data leakage.

Fold 01

Train: EuroBasket 2005

-> Test: FIBA World Cup 2006 (Out-of-sample)

Fold 05

Train: 2005–2009 Tournaments

-> Test: FIBA World Cup 2010 (Out-of-sample)

Fold 11

Train: 2005–2016 Tournaments

-> Test: FIBA EuroBasket 2017 (Out-of-sample)

Fold 17

Train: 2005–2023 Tournaments

-> Test: Paris Olympic Games 2024 (Out-of-sample)

Why Walk-Forward Matters: Random K-Fold leaks modern 3-point volume into past eras.

Monte Carlo Tournament Simulation

180,000 full-bracket simulations combining calibrated probabilities and ratings.

Dynamic Bracket Flow

Simulates group stages, tie-breakers, and knockout brackets accounting for match sequence and cumulative fatigue.

Shrinkage toward Mean

Applies Bayesian shrinkage to pre-tournament ratings to prevent over-reacting to preliminary round upsets.

Scenario Sensitivity

Enables coaching staff to test 'what-if' roster changes or key injury impacts before tournament commencement.

Output: Probabilistic medal probabilities and conditional matchup trees.

DECISION SUPPORT

Interactive Workspace & Anti-Hindsight Mode

Isolating pre-game evidence states at T-30, T-7, T-1, and Game Day to eliminate hindsight bias.

Temporal Quarantine

The system hides post-game outcomes, forcing the analyst to evaluate the exact pre-game information available before tip-off.

Multi-Layer Evidence Matrix

Synthesizes Four Factors, shot zones, archetype matchups, and predictive ratings in one unified view.

Automated Inconsistency Alerts

Flags statistical red flags (e.g. low-sample shooting spikes vs career priors) before they reach the coach.

Interactive Tool: Streamlit workspace runnable locally with `streamlit run`.

03

Selected Case Studies

Four Applied Demonstrations: Tactical Support, Data Engineering, ML Rigor, and Longitudinal R Analysis

From Data to the Whiteboard: Beijing 2008 Final

How pre-game analytics framed the tactical strategy for Spain vs. USA 'Redeem Team'.

Quantitative Pre-Game Signals

- Context: Spain lost by 37 pts (82-119) in group phase against USA.
- Half-Court Signal: In 5v5 half-court sets, Spain held a +4.2 Net Rating advantage via Gasol brothers inside.
- Transition Vulnerability: USA scored 1.25 PPP on transition off turnovers.
- Defensive Flaw: USA bigs dropped deep into paint to pack driving lanes.

Tactical Whiteboard Brief

- Pace Control: Limit total possessions below 72 using 16+ second attacks.
- Transition Defense: Implement 2-3 matchup zone immediately after made baskets.
- Pick-and-Pop Spacing: Exploit deep drop with 3-pointers from Pau, Marc, and Garbajosa.
- Court Outcome: Spain cut deficit to 4 pts (108-104) at 2:20; final 107-118.

Demonstration: Turning complex data into 3 concrete tactical questions for the head coach.

In-Process OLAP Lakehouse with DuckDB & Parquet

Deterministic data engineering for 20 years of heterogeneous international tournaments.

Engineering Challenges

- Heterogeneous FIBA Boxscores across 2005-2024 with varying column conventions.
- Diacritic Name Variations across international alphabets (Nowitzki, Spanoulis, Bogdanovic).
- Zero Cloud Infrastructure cost constraint: 100% offline local reproducibility.
- Dual-Language Access: Zero-copy querying from both Python and R.

Architecture Solutions

- DuckDB Relational Engine: 12 tables, Star Schema, in-process ACID transactions.
- 11 Apache Parquet Marts: Snappy-compressed columnar storage with SHA-256 hashes.
- Deterministic Entity Resolver: 2,124 canonical player IDs without paid commercial APIs.
- Automated QA Suite: 227 tests in pytest enforcing 200 min/game invariants.

Demonstration: Production-quality data engineering capable of scaling to domestic leagues.

Zero-Leakage ML with 17 Chronological Folds

Avoiding temporal leakage and delivering calibrated probabilities for tournament play.

Methodological Traps Avoided

- Random K-Fold Cross-Validation leaks future playstyle evolution into past evaluations.
- Over-confident predictions: raw neural nets produce uncalibrated probabilities in short events.
- Overfitting on small samples: complex models over-index on single-game tournament anomalies.
- Causal Overclaiming: confusing feature importance with tactical ground truth.

Rigorous ML Protocol

- Expanding Window: 17 folds evaluating strictly unseen tournaments (1,105 test games).
- LightGBM with L2 Regularization on pre-game Four Factors differentials.
- Isotonic Probability Calibration: Brier Score 0.1967 (vs 0.2500 naive), ECE 0.0314.
- TreeSHAP Attribution: Interpretable local feature contribution without causal exaggeration.

Demonstration: Scientific honesty and calibrated probability estimation.

Longitudinal Shooting & Role Mining in R / Quarto

Multi-year player evaluation, non-parametric bootstrap, and reproducible reporting.

Statistical Analysis in R

- Direct DuckDB Connection via DBI: Zero data duplication between Python and R.
- Longitudinal Career Tracking: 3,767 qualified player campaigns (>=40 minutes).
- Cluster Bootstrap (B = 5,000): Non-parametric 95% confidence intervals for True Shooting %.
- Rebounding & Turnover Stability: Higher year-over-year correlation (r > 0.65) than 3P%.

Clustering & Quarto Reporting

- PCA Decomposition: First 3 principal components explain >60% of player variance.
- K-Means++ Clustering: Discovery of 6 functional archetypes based on 14 normalized metrics.
- Quarto CLI Pipeline: Automated compilation into interactive HTML report with ggplot2.
- Custom Visualization: `theme_basketball_analytics()` with 300 DPI vector exports.

Demonstration: Advanced R, tidyverse, statistical inference, and reproducible reporting.

04

Validation & Engineering Rigor

227 Automated Pytest Tests, Cross-Language Verification, and Deterministic Master Runner

ENGINEERING QUALITY

227 Automated Regression Tests (100% Pass)

Continuous integration test suite covering 26 specialized test modules.

Data Integrity & Invariants

Validates 200 min/game, boxscore closure, SHA-256 hashes, DuckDB schemas, and column contracts across all 18 tournaments.

Analytical & ML Precision

Tests walk-forward temporal splits, Brier Score bounds, ECE limits, Monte Carlo distribution bounds, and SHAP attribution coherence.

Portfolio & Cross-Language

Verifies R-DuckDB parity, Quarto compilation, publication package completeness, and zero prohibited marketing buzzwords.

Execution Speed: Full suite executes in ~2 minutes on standard hardware.

INTEROPERABILITY

Cross-Language Parity: Python + R + DuckDB

Ensuring identical analytical results regardless of execution environment.

Unified Warehouse	Single `basketball_analytics.duckdb` file accessed concurrently by Python (`duckdb`) and R (`DBI::dbConnect`).
Zero Discrepancy	Exact match on 1,145 games, 2,290 team rows, 27,353 player boxscores, and 4,350 qualified records.
Metric Synchronization	Identical eFG% (0.5355), Pace (61.72), and 3P Attempt Rate (0.3153) across Python and R pipelines.
Decoupled Design	Python handles ETL/ML; R handles statistical inference and visualization; Parquet provides columnar handoff.

Verified: Deterministic script `scripts/verify_cross_language.py` checks parity automatically.

ONE-COMMAND EXECUTION

Deterministic End-to-End Reproducibility

Run the entire project verification pipeline with a single master command.

```
$ python scripts/run_project.py
```

1. Environment Check

Verifies Python 3.10+, R 4.4+, DuckDB, PyArrow, LightGBM, Pandas, and Pytest.

2. Warehouse Check

Validates DuckDB database (16.8 MB) and 11 Parquet analytical marts.

3. R Analysis Pipeline

Executes R statistical scripts and generates figures in reports/figures_r/ in ~21s.

4. Pytest Regression Suite

Runs all 227 automated tests with 100% pass rate in ~2 minutes.

Zero Live Dependencies: Everything needed is packaged inside the repository.

05

Limitations & Professional Scope

Scientific Honesty, Transparent Boundaries, and Operational Value for Basketball Clubs

Transparent Boundaries: What the System Claims

Clear distinction between data-backed capabilities and operational limits.

What the System DEMONSTRATES

- Pace-neutral Four Factors and True Shooting evaluation.
- Longitudinal Bayesian shrinkage to control short-tournament variance.
- Strict out-of-sample calibrated match probability modeling.
- Structured pre-game briefs that ask the right tactical questions.
- Fully reproducible in-process OLAP data lakehouse.

What the System DOES NOT Claim

- No 25Hz live optical tracking (no Second Spectrum camera telemetry).
- No automatic game decisions: data structures evidence, coach decides.
- No infallible betting prediction: models estimate uncertainty.
- No causal guarantees: statistical associations are not mechanical laws.
- No real-time in-game bench tracking (focused on pre/post game).

Scientific Integrity: Honest analytics earns the trust of coaching staffs.

ORGANIZATIONAL IMPACT

Operational Value by Organizational Role

How different stakeholders in a basketball club extract value from this platform.

Head Coach & Assistants

Saves 15+ hours/week by synthesizing rival scouting into 1.5-page pre-game briefs with Four Factors and P&R drop alerts.

Sporting Director & Scouts

Filters out small-sample shooting noise with Bayesian shrinkage and audits roster complement via 6 objective functional archetypes.

Analytics Lead & CTO

Provides a zero-cloud-cost, 227-test verified OLAP foundation in DuckDB and Parquet, ready for domestic league integration.

Design Focus: Built for immediate operational adoption in professional basketball environments.

Day-1 Club Onboarding Roadmap

How this system adapts to domestic club leagues (ACB, EuroLeague, BBL, NBA).

Days 1–10: Data Ingestion

Connect domestic league boxscores (Genius Sports, Synergy, EuroLeague API) into the DuckDB raw staging layer.

Days 11–20: Metric Customization

Calibrate Four Factors baselines and Bayesian priors for the domestic league's specific pace and refereeing trends.

Days 21–30: Staff Integration

Deliver automated weekly pre-game briefs directly to coaching staff and train video coordinators on inconsistency alerts.

Ready to Deploy: Modular architecture designed for seamless data source substitution.

CONCLUSION

Summary & Key Takeaways

A disciplined, reproducible approach to basketball quantitative analytics.

1. Data-First & Rigorous

20 years of FIBA tournaments structured in DuckDB and Parquet with 227 automated tests and zero data leakage.

2. Calibrated Machine Learning

Brier Score 0.1967 and ECE 0.0314 across 17 chronological folds (1,105 test matches).

3. Actionable Tactical Impact

1.5-page briefs that ask concrete questions for the whiteboard instead of dumping raw statistical charts.

4. Transparent & Reproducible

Complete open-source pipeline executed with a single command on GitHub.

Core Philosophy: Data structures evidence. The coach decides the game.

PROJECT REPOSITORY & CITATION

International Basketball Analytics

Explore the Full Open-Source Codebase, Case Studies, and Automated Tests

GitHub Repository: https://github.com/miguelo0203/basketball-analytics

Release v1.0.0 (Data-First, Verified, MIT License)

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